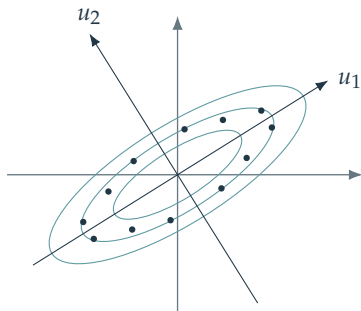


### Data covariance

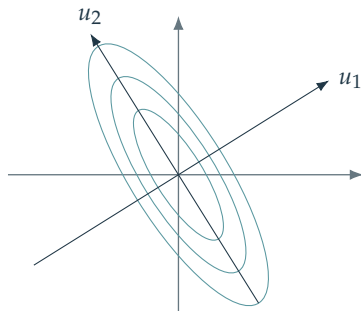
Width along  $u_i \propto \sqrt{\lambda_i}$



$$C/n = U \operatorname{diag}(\lambda_i) U^\top$$

### Quadratic level sets

Width along  $u_i \propto 1/\sqrt{\lambda_i}$



$$\langle Cx, x \rangle / n = \text{constant}$$

Semiaxes:  $r\sqrt{\lambda_i}$  (left),  $r\sqrt{\lambda_1\lambda_2}/\sqrt{\lambda_i}$  (right),  $r = 1, \frac{3}{2}, 2$ .